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A silicon continuous position sensitive diode and associated electronics: modelling and simulation

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Abstract

In contrast to the segmented detectors (micro-strips and pixels) widely used in high energy physics experiments, continuous detectors such as position sensitive diodes (whose localisation principle is based on charge division) remain of interest when count rate and multiplicity are not critical. That is the case with several imaging applications in nuclear medicine and biology. In that context, our group has developed a γ camera dedicated to the detection of body tumour tissue during operative cancer surgery. This per-operative compact imager consists of an image intensifier tube viewed by a PSD of $25 \times 25 \text{ mm}^2$. In order to improve detection performances of this camera we have developed a simulation model of the PSD (schematic form) and associated electronics (behaviour form) using CAO AWB/Cadence. This allows an accurate estimate of both spatial resolution and linearity. These simulations provide an exhaustive study of the performance of various types of PSD and readout electronics configurations. In our application which requires high resolution imaging, we have demonstrated that a classical detector filtered by a trapezoidal shape offers the best compromise between linearity and spatial resolution. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

High energy physics experiments require large equipments, which reach a high multiplicity (large number of particles should be detected per events) at high counting rates. For these reasons, segmented detectors like specific silicon micro-strip and now pixel detectors are widely developed and implemented for tracking [1]. However, in some applications like nuclear med-

icine where multiplicity and counting rate are not critical, continuous Position Sensitive Detectors (PSD) are considered a relevant solution to achieve performances suited to medical diagnosis or radio pharmacological study on small animal models [2].

The first PSD development has begun in the sixties, before segmented detectors. Continuous PSD are actually available in standard forms for surfaces up to $60 \text{ mm} \times 60 \text{ mm}$, four or five standard electronic channels are enough to read-out the whole surface of detection. These characteristics guide us to implement a PSD to develop

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a hand-held per-operative γ camera dedicated to inside body localization of radio-labelled tumours during cancer surgery. This device consists of a head module composed of a high-resolution tungsten collimator and a scintillator crystal readout by an image intensifier tube optically coupled with a PSD. The spatial performances of the compact camera are strongly influenced by PSD geometry and associated electronics. To choose the optimum configuration, simulation and analytic calculations to date have principally dealt with one-dimensional PSD. In order to obtain optimum spatial performances of two-dimensional PSD, we have developed a simulation model including detectors and associated electronics that allows to accurately simulate both spatial resolution and linearity.

In this paper, we recall briefly the principle of continuous PSD and we report theoretical models of the spatial resolution including both noisy and distortion simulations. Results obtained for two types of detectors and several electronic filters will be compared.

2. Method and model

2.1. Principle

Continuous PSD is a fully depleted p + N diode (300 μm thick) of 50 $\times$ 50 mm² area. Its back side (N side) is formed by a metallic cathode and the front surface (P + implantation side) is a resistive sheet ($R_{\square} = 10\text{--}22 \text{ k}\Omega/\text{sq.}$). Currents signals are obtained from the four corners of the square resistive electrode and from the back side (Fig. 1).

When the centre point of the PSD is set at the original point, the position measurement is given by

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{L}{2} \frac{1}{Q_E} \begin{pmatrix} Q_{X1} - Q_{X2} \\ Q_{Y1} - Q_{Y2} \end{pmatrix} \quad (1.1)$$

where L is the PSD dimension, $Q_{X1} = Q_C + Q_D$, $Q_{X2} = Q_A + Q_B$, $Q_{Y1} = Q_A + Q_C$, $Q_{Y2} = Q_B + Q_D$, where Q_I ($i = A, B, C, D$) are inductive charges. Q_E (normally $= Q_A + Q_B + Q_C + Q_D$),

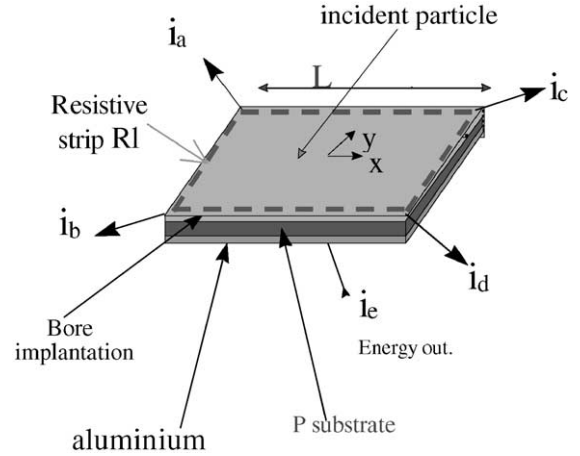


Fig. 1. The two-dimensional position is given by the following hypothesis: $i_a + i_b = i_c(L/2 - x)/L$, $i_c + i_d(L/2 + x)/L$, $i_a + i_e = i_c(L/2 - y)/L$, $i_b + i_d = i_c(L/2 + y)/L$ (the origin is at the centre of the photodiode). The low resistive strip diffused around the detector reduces classical distortion figure but introduces an additional thermal noise.

the backside's inductive charge, is proportional to the total energy deposited. Corrected PSD also includes a border strip to reduce distortion (Fig. 1). R_l is the resistance of this strip per unit of length. In order to have the best position linearity, the product LR_l should be equal to or less than $R_{\square}/10$ [3].

2.2. Spatial resolution: theoretical approach

Spatial resolution is given by taking variances $V(x)$, $V(y)$ of functions (1.1) [4]. The statistical fluctuations (dQ_i) at the four channels are independent and equal. As $Q_E = \sum Q_i$ (current conservation):

$$V(x) = L^2 \frac{1 + (x/L/2)^2}{Q_E^2} dQ_A^2 \quad \text{same for } y. \quad (1.2)$$

It can be observed that the fwhm spatial resolution ($\delta(x) = 2.355\sqrt{V(x)}$) is linked to the position itself (optimal resolution is at the centre and is increasing at a $\sqrt{2}$ factor close to the edges). Spatial resolution is also proportional to the PSD length and inversely proportional to the energy deposited (this state is perfectly verified by experiment). As a

consequence, we can predict the $\sqrt{dQ_i^2}$ required to achieve a spatial resolution at a given energy.

2.3. Equivalent noise charge

since both PSD dimension and particle energy are generally fixed, the spatial resolution could be improved by decreasing the Equivalent Noise Charge ($ENC = \sqrt{dQ_i^2}$). Fig. 2 shows the theoretical model classically used for the detector and the front end electronics to express the ENC.

The rms noise (in Coulomb) at the input (main contribution) is:

$$ENC = \left(\frac{aC_{Det}^2}{\tau} a_{F1} + b\tau a_{F2} + \frac{a}{R_{Det}^2} \tau a_{F2} + \frac{2kT}{R_{Det}} \tau a_{F2} \right)^{1/2}.$$

The resistive layer implies also a “doublet” of the series noise “a” and a $2kT/R_{Det}$ thermal noise. Non-dimensional form factors $a_{F1,2}$ are given by integrations of the weighting function [5].

2.4. Distortion

The intrinsic inductive charge signals are time and position dependant: $Q_i = Q_i(t, x, y)$. The spatial non-linearity depends on the filter’s shape and on the relative values of τ (shaping time) and τ_{Det} . ($= C_{Det} R_{Det}$). A ballistic deficit occurs when τ is closer to $\tau_D/2$ [5]. To take account of the ballistic deficit effect on spatial resolution, we have to introduce the effective collected charges

Q_i in formula (1.2):

$$\begin{aligned} \left(\frac{\delta(x)}{\delta(y)} \right) &= 2.355 \frac{LENC}{\left(\sum_{i=A}^D Q_i \right)^2} \sqrt{2} \\ &\times \left(\frac{\sqrt{(Q_C + Q_D)^2 + (Q_A + Q_B)^2}}{\sqrt{(Q_B + Q_D)^2 + (Q_A + Q_C)^2}} \right). \end{aligned} \quad (1.3)$$

One can note that the spatial resolution is improved by taking $\sum Q_i$ from the back side (no deficit).

Due to the detector time constant, the spatial resolution and linearity are linked. The compromise is to implement a shaping filter with short time constant for good spatial resolution and to have a sufficiently large time constant to avoid picture distortion. The effective distortion and spatial resolution are not analytically computable in any case. Electronics simulation is an elegant solution to take into account all effects (2-dimensional PSD, pin cushion geometry, $R_{\square}$, R_l , distributed C_{Det} , ballistic deficit, non-linearity, filters) and to find the best trade-off.

3. Simulation model

3.1. PSD

A 13×13 cells RC array models the detector (Fig. 3) (a 11×11 cells elements configuration is just enough to achieve a realistic spatial linearity [3]). Each basic cell consists of an elementary capacitance $C_{elem} = C_{det}/169$ connected to the four neighbours via four resistances $R_c = R_{\square}$, where C_{det} is the total junction capacitance [3]. In order to simulate properly the detector’s borders effect and the resistive strip, cells placed along the edge exhibit particular characteristics: $C_{elem} = C_{det}/338$ and $R_B = |R_l|/12$ for a corrected detector. The typical parameters are $C_{Det} = 900$ pF, $L \times R_l = 1.8$ k Ω and $R_{\square} = 18$ k Ω . A current pulse could be injected at each C_{elem} boundary. Pulse is 15 ns times larger corresponding than the charge drift time.

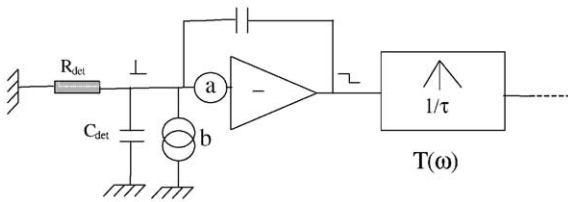


Fig. 2. A classical front end electronic is connected to each corner’s pad. R_{det} is the resistor viewed by an electronic (each corner is a virtual ground). C_{det} is the total detector’s junction capacitance “a” and “b” are bilateral series (in V²/Hz) and parallel noise (in A²/Hz), respectively, $T(\omega)$ is the shaping amplifier’s transfer function of time constant τ .

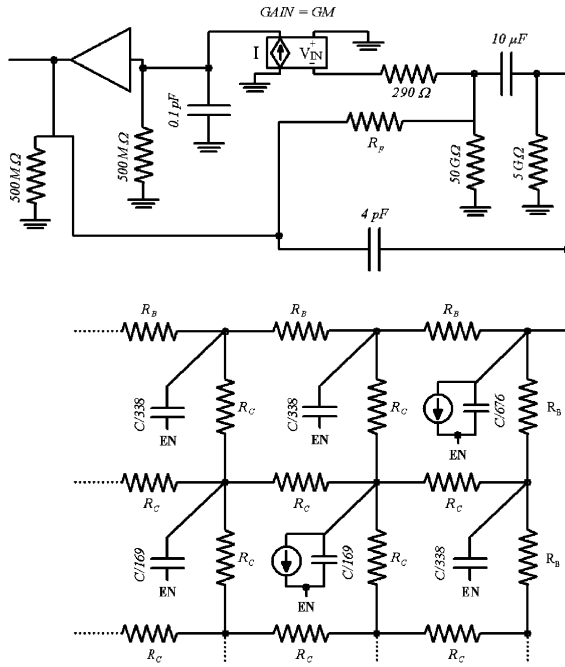


Fig. 3. The detector is a 13×13 matrix. Each cell is an elementary capacitance connect to the neighbouring cells via a resistor $R_c = R_{\square}$. For a corrected detector, the resistors at the border are $R_B = L \times R_{\square}/12$. The pre-amplifier (quite perfect) has a 4 pF feed back capacitance.

3.2. Charge sensitive pre-amplifier

In practice, we note a discharge of the feed-back capacitance of the pre-amplifier via the resistive layer of the PSD. However, to study the PSD and the electronic filters intrinsic performances, the charge sensitive pre-amplifier is designed quite perfectly as a voltage sensitive current source with high gm followed by a unity gain buffer (Fig. 3).

3.3. Filters

Elementary Laplace blocks from “Profile Behaviour” library of AWB implement electronic filters. These are followed by sample and hold peak detection realised by programming function blocks from the same library. Laplace

modules accept both time and frequency Spice simulation domain. Noise analysis results are given as $10 \log V_{\text{rms}}^2$ versus frequency and an AWB’s function computes directly the total noise of the bandwidth. The amplifier gain is adjusted (in time domain) to have a 1 V response when using an α particle pulse (calibration in charge).

3.4. Noise model

In order to verify that AWB-Cadence’s simulation and ENC calculation are in close agreement, the simple test circuit corresponding to the simplified model of PSD and electronics front end was used (Fig. 2). The thermal noise of R_{det} is the main contribution to the parallel noise, where R_{det} is the resistance viewed by the electronics. In that case, the (parallel) noise “b” associated to the leakage current is negligible. An explicit Johnson noise resistor R_s placed at the input of the pre-amplifier is modelled on the serial noise (Fig. 2). Five electronic filters are evaluated: RC–CR, Gaussian, trapezoidal (with two different parameters “a”) and gated integrator (with rectangular pre-filter). Timing methods are also simulated (propagation delay, fixed threshold, rise time and constant fraction discrimination).

The Gaussian filter is formed by an RC–CR followed by 6 first order integrations (CR–RC7). The trapezoidal shaping is implemented by three delay lines followed by a perfect integrator [6]. For the noise test, the width’s sloped part is τ (integration) and the flat part is $a \cdot \tau$ ($a = 0.4$ and 4.0) [6]. For the gated integrator filter (time variant filter), the weighting function is different from the transfer function [6,7]. Therefore, two different Spice schematics are required: one in time domain (spatial non-linearity analysis) and a second in frequency domain. In the first time domain, we used the gated integrator of width τ preceded by a rectangular pre-filter of width τ [7]. In that condition, the corresponding weighting function is the well known triangular shape of base 2τ [7] implemented with delay lines [10].

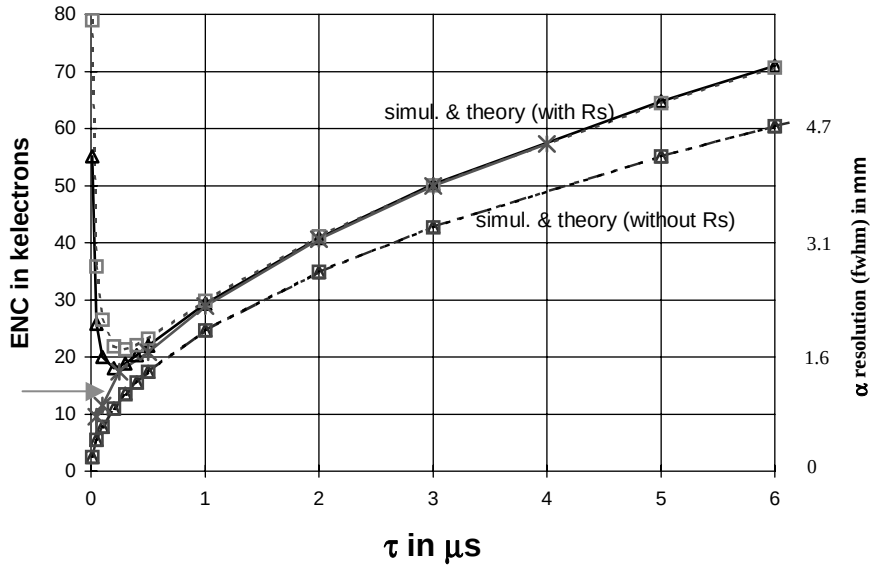


Fig. 4. Simulation performed on the classical test circuit taking into account serial and doublet noise (with and without R_s curves). One can note that results are in good agreement with ENC calculation (theory curves). This allows extrapolating methodologies to complex detectors (with two-dimensional distributed RC). Curve with arrow is the simulation results obtained with the distributed detector: the serial noise is then completely negligible. Spatial resolution at the centre of the detector is reported on the right axis.

4. Results

4.1. Noise

The ENC results for the test circuit (simplified detector model) (Fig. 2) and the complete detector (Fig. 3) are reported in Fig. 4. We can note that frequency noise analysis of the simulated test circuit (with or without R_s curves Fig. 4) gives similar results to the ENC calculation (theory curves) and perfectly takes into account serial and doublet noise (difference between with and without serial noise R_s). More generally, this simulation allows characterising absolute performances of each filter. For example, the ratio between the optimum ENC obtained with the RC–CR ($\tau = 300$ ns) and the CR–RC7 (at $\tau = 100$ ns) is 1.17, while the theory gives 1.16. For the trapezoidal shaping simulation, results also fit perfectly the theoretical formula: $ENC_{||} = (\frac{2}{3}\tau + a\tau)$ [6,8].

In opposition to the classical model (test circuit Fig. 2) commonly used, we can see in Fig. 4 (curve

with arrow), when the complete detector is assumed (2 dimensions), that the capacitance C_{Det} does not affect the ENC (in fact, the ENC does not increase when τ decreases). For a one-dimensional detector, the capacitance viewed by an electronic channel is $C_{Det}/3$ at low frequency [5]. In any two-dimensional detector's pincushion, this viewed capacitance is considerably more reduced and the effect of the apparent distributed capacitor on the noise becomes completely negligible. However, the capacitance affects the time constant of the signal pulse induced by an incident particle.

The simulation of our corrected detector also confirms that the main noise contribution is the Johnson noise of the resistive layer. Consequently, in order to optimise the energy resolution ($= ENC^2 \cdot 2.355 \cdot 3.615$ eV), the spectroscopy (discrimination by energy deposited) should be done at the back side (metallic layer). As the parallel noise contribution becomes dominant for any filter, τ should be as small as possible so as to improve spatial resolution. To estimate this resolution, the

computed ENC was injected in formula (1.3) that takes into account the position itself and the ballistic deficit. The values obtained at the centre of the detector (with 5.5 MeV α particle) are reported on the right y axis of Fig. 4.

4.2. Distortion

CSA signals depend on the incident particle position and on the intrinsic time constant $\tau_{\text{Det}} = R_{\square} C_{\text{Det}}$ corresponding to the detector characteristics. When the amplifier shaping time constant is

reduced, the charge collection is not completed and a spatial distortion called ballistic deficit is introduced. Fig. 5 shows the pictures obtained for two time constants and for both types of detector. Intrinsic distortions (large τ) are shown on the left pictures and the distortion figures due to the ballistic deficit are shown on the right. These distortion figures have been confirmed by measurements (epoxy mask irradiated by α particles) that also allowed adjusting the simulation parameter $R_{\square}$. As we see in Fig. 5, the classical detector exhibits a good spatial resolution but

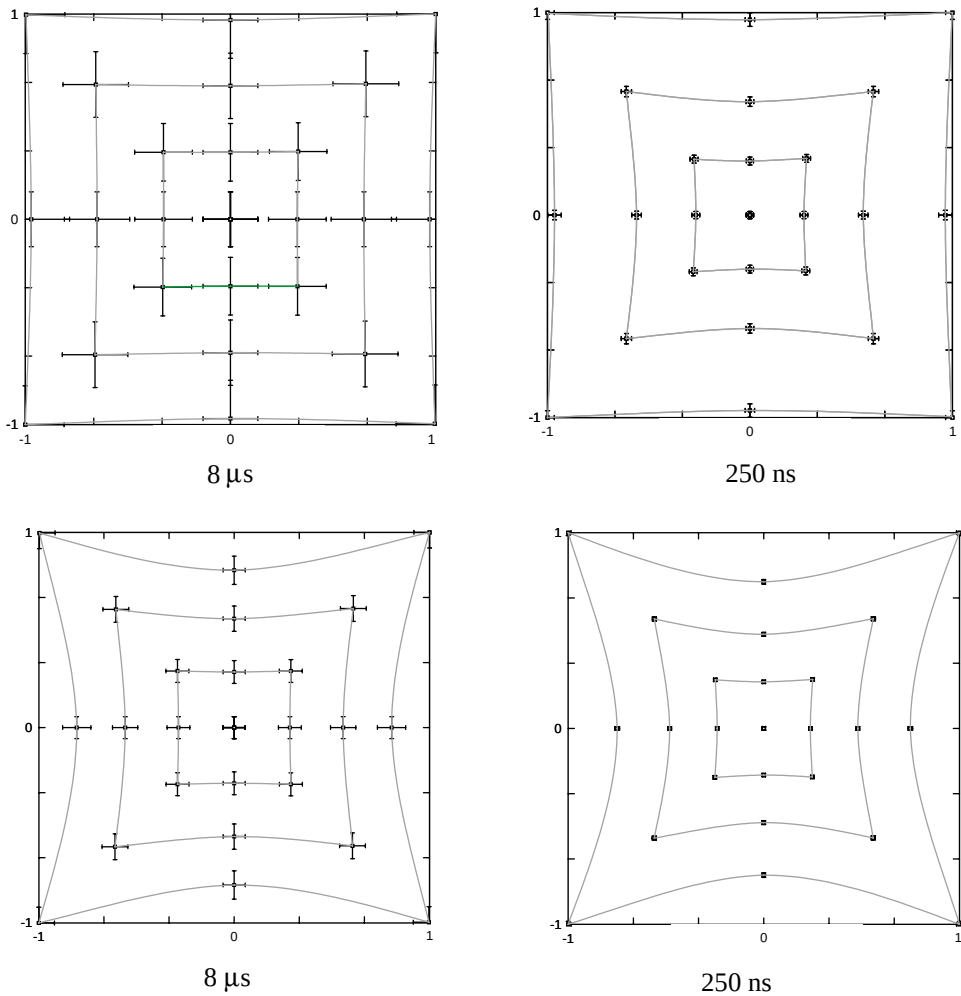


Fig. 5. Distortion figures. Both upper pictures are relative to a corrected detector for $\tau = 8 \mu\text{s}$ (intrinsic distortion) and $\tau = 250 \text{ ns}$ (ballistic deficit). Both lower pictures concern a classical detector. For each point, the resolutions δx and δy are represented by error bars.

poor spatial linearity. On the contrary, the intrinsic localisation of the corrected detector is fine but its spatial resolution is not acceptable, this is due to the low value of R_{Det} . Indeed, in this kind of detector, the resistor viewed by an electronic channel is $R_{\text{Det}} \approx L R_1/2 \approx 980 \Omega$ (two strips in parallel) that induces a high thermal noise proportional to $\sqrt{2kT/R_{\text{Det}}}$.

4.3. Discussion

We have shown that a compromise between distortion (high value of τ) and position resolution (small τ) must be reached. The calculated versus expected position along y -axis is plotted in Fig. 6 for both types of detector (at $\tau = 100$ ns). By definition, the distortion (non-linearity) value is the maximum difference between computed positions and the regression line (and not the expected positions). A linear operation consists of multiplying the position found by a scale factor (picture's magnification) which is the inverse of the fit line slope. This linear operation must be done in the same way for both positions and associated spatial resolutions. To simplify, the correction is performed in one dimension, along y -axis (same as x -axis) and along the diagonal.

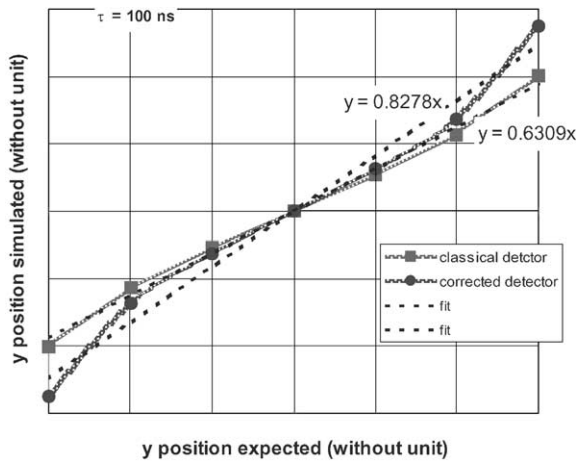


Fig. 6. Non-linearity for both detector at $\tau = 100$ ns (ballistic deficit mode). Points are located along the detector y -axis. In ideal case, points should be on the first diagonal. Distortion should be calculated with respect to linear fit.

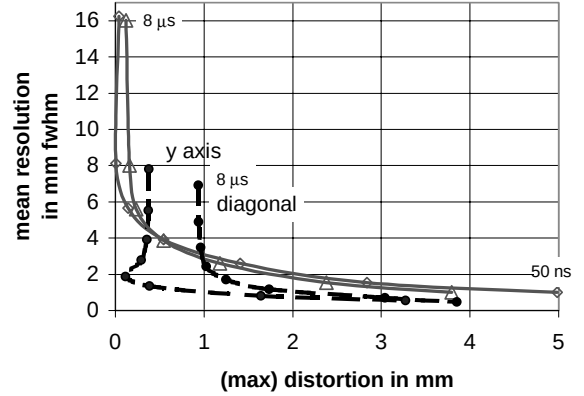


Fig. 7. Mean spatial resolution versus distortion for both type of detectors (dashed curves for classical detector) and for different time constants. Both resolution and distortion include the linear correction. At 1 mm distortion and 3 mm resolution (in α particle) both detectors are equivalent.

Fig. 7 shows the mean spatial resolution (with the linear correction applied) as a function of distortion (including the magnification $1/\text{slope}$). Both detectors can achieve a 1 mm distortion corresponding to a 3 mm fwhm resolution. This distortion can be better than 1 mm with the corrected PSD but the spatial resolution becomes rapidly poor. For our application, the classical detector achieves an acceptable resolution (with a distortion more than 1 mm). More generally, the choice of the detector also depends on whether or not we accept a non-linear 2-dimensional image treatment. Both detectors' slopes are not weak in allowing the possibility of an efficient correction. In that context, the simulation provides the distortion figures' data to perform the image treatment (independent of the experimental spatial resolution).

The effect of shaping filters on spatial resolution and distortion is considered in Fig. 8. The gated integrator (time-variant filter) is not a good candidate due to its pre-filter, which is rectangular (no derivation). A pre-filter based on derivation will normally provide better results. The trapezoidal filter ($a = 4.0$) provides less distortion than Gaussian filter with the same noise. ($a = 4.0$ was founded by simulation). However, one can note that if we accept a distortion greater than 2 mm

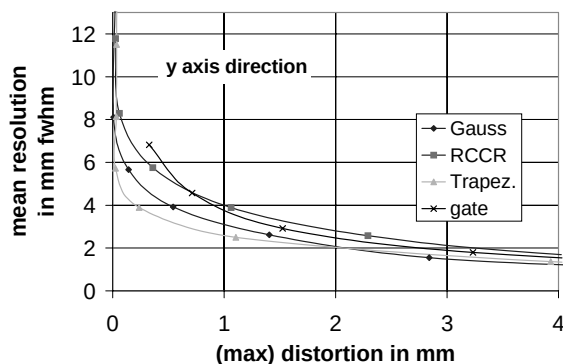


Fig. 8. Mean spatial resolution versus maximum distortion for different τ . The trapezoidal ($a = 4.0$), which has the same ENC as the Gaussian, provides less non-linearity because the deficit ballistic is reduced.

(towards y -axis) and 1.4 mm (towards diagonal direction), than the Gaussian filter becomes the better choice.

The position reconstruction simulated by all timing methods gives quite linear responses, while the timing noise (amplitude_noisy/slope) is too high. Since the Johnson noise is important compared to the signals' weak slopes (specially far from the centre), the timing resolution and consequently the spatial resolution rapidly becomes poor.

5. Conclusion

We have developed a simulation model to predict absolute distortion and spatial resolution performances of position sensitive diodes and its associated electronics. The good agreement between simulations, theoretical and experimental results demonstrated that this model is a powerful tool that can be used to study various kinds of detectors and electronics. AWB/Spice also allows

to include secondary effects, like real electronics, by introducing a schematic design with noisy sources to study non-ideal charge pre-amplifier and filters.

Simulations give absolute results that allow comparing different PSD and electronics. For our medical application where the spatial resolution is required, we have demonstrated that a classical detector readout by a trapezoidal shape ($a = 4$) offers the best trade off between spatial resolution and position linearity.

Further studies are already scheduled such as the trapezoidal filter's performances optimisation of the parameter " $a\tau$ ". This filter will be compared to the gated integrator with derivation pre-filter, which could be a good candidate [9,10]. Furthermore, a duo lateral detector type (two resistive layers on both sides) will be simulated. This photodiode reduces the distortion but has poor spectroscopy resolution. Other fundamental questions will be addressed, like the optimisation of R_{Det} and the studies of a pre-amplifier suitable to the resistive PSD impedance.

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